

Windows Scheduled Task Persistence — Attack Simulation & Incident Response Lab Write-Up

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Lab Type: Home Lab / Incident Response Investigation

Environment: Kali Linux → Windows 10

Focus: Windows Endpoint Detection, Investigation, Persistence & Incident Response

1. Executive Summary

This lab was designed to simulate a Windows endpoint compromise and demonstrate the investigation process from an incident responder's perspective.

An isolated Windows environment was configured with **Sysmon** and Windows Security auditing to provide endpoint telemetry. A simulated attacker then delivered and executed a suspicious executable, established persistence through a Windows scheduled task, and generated network activity.

The investigation was conducted using **Windows Event Viewer**, **Sysmon**, and **Windows Security event logs**.

The primary evidence used during the investigation was:

- Sysmon Event ID 1 — Process Creation
- Sysmon Event ID 3 — Network Connection
- Windows Security Event ID 4698 — Scheduled Task Creation

The investigation resulted in a **True Positive** determination based on the correlation of process execution, network activity, and scheduled-task persistence.

The original lab defines the objective as simulating a reverse shell, establishing scheduled-task persistence, and then investigating the resulting activity using Sysmon and Windows Security logs.

2. Lab Objectives

The objectives of this lab were to:

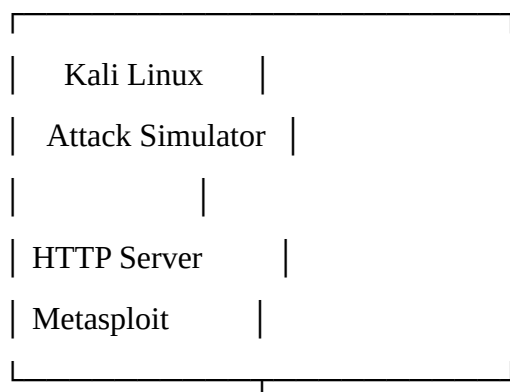
1. Build an isolated attacker/victim environment.
2. Configure Windows endpoint logging.
3. Simulate suspicious executable execution.
4. Investigate process creation.
5. Investigate outbound network connections.
6. Identify scheduled-task persistence.
7. Correlate multiple event sources.
8. Build a chronological incident timeline.

9. Identify Indicators of Compromise (IOCs).
10. Map observed behavior to MITRE ATT&CK.
11. Produce a professional incident-response report.

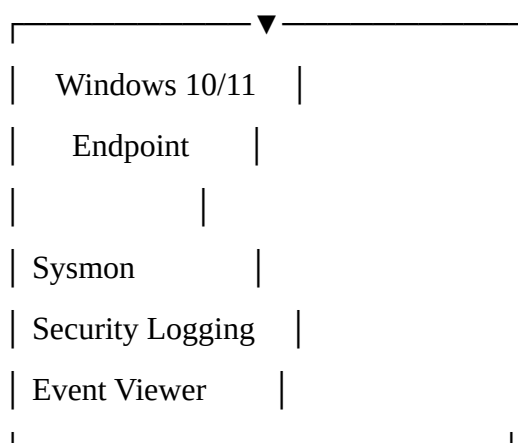
3. Lab Architecture

The environment consisted of two virtual machines.

Isolated Lab Network



Host-Only Network



4. Tools Used

Tool	Purpose
VirtualBox	Virtualization
Kali Linux	Attack simulation
Windows 10/11	Investigation endpoint
Sysmon	Endpoint telemetry
Windows Event Viewer	Log investigation
Windows Security Logs	Security event investigation
Metasploit	Controlled attack simulation
HTTP Server	Controlled file-delivery simulation
MITRE ATT&CK	Technique classification

5. Windows Logging Configuration

Before conducting the simulation, Sysmon was installed on the Windows endpoint.

Sysmon was selected because it provides detailed telemetry relating to:

- Process creation
- Network connections
- Process relationships
- File and system activity

The Sysmon logs were available under:

Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational

Windows Security auditing was also configured to capture scheduled-task creation events.

The lab specifically identifies **Security Event ID 4698** as the event used to investigate scheduled-task creation.

6. Attack Simulation

6.1 Suspicious Executable

As part of the controlled simulation, a suspicious executable was introduced to the Windows endpoint.

The executable was placed in:

C:\Users\Public\Windows11InstallationAssistant.exe

The purpose was to simulate a suspicious program being introduced to an endpoint and subsequently executed.

The use of a legitimate-looking filename demonstrates how an attacker might attempt to make malicious activity appear less suspicious.

7. Initial Execution Investigation

After the executable was executed, I investigated **Sysmon Event ID 1 — Process Creation**.

The investigation focused on identifying:

- Executable path
- Process name
- Command line
- Parent process
- User
- Timestamp

Event ID: 1

Image: C:\Users\Public\Windows11InstallationAssistant.exe

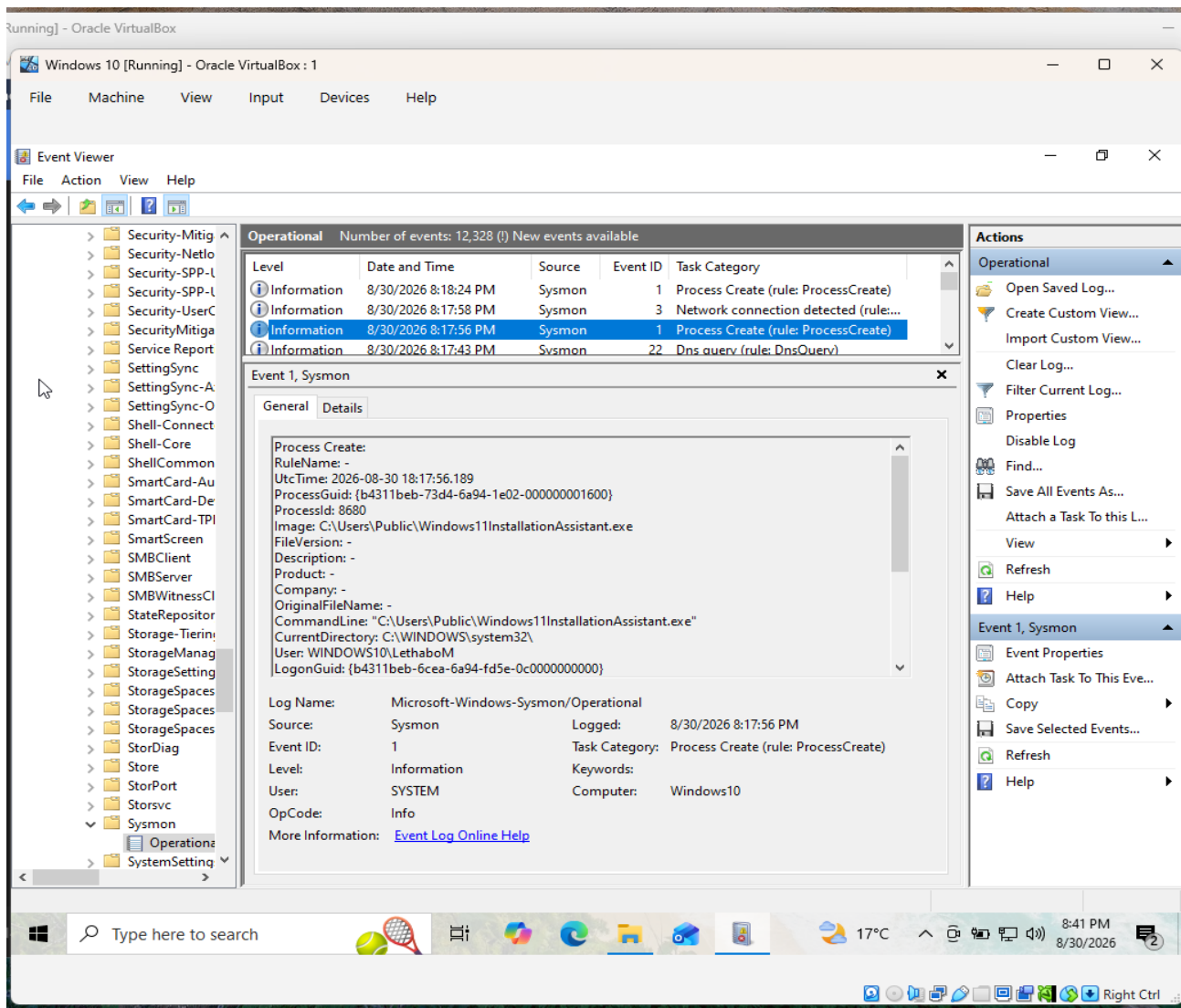
Timestamp: 30 August 2026 08:17:56 PM

User: Windows10\LethaboM

Command line: "C:\Users\Public\Windows11InstallationAssistant.exe"

Evidence

Screenshot 01 — Sysmon Event ID 1



Finding

The event demonstrated that the suspicious executable was not merely present on the system; it had actually been executed.

This established the first important stage of the investigation:

Suspicious file → Execution

8. Network Connection Investigation

I then investigated **Sysmon Event ID 3 — Network Connection**.

The objective was to determine whether the suspicious process communicated with another system.

The relevant fields included:

Event ID: 3

Image: C:\Users\Public\Windows11InstallationAssistant.exe

Destination IP: 192.168.0.104

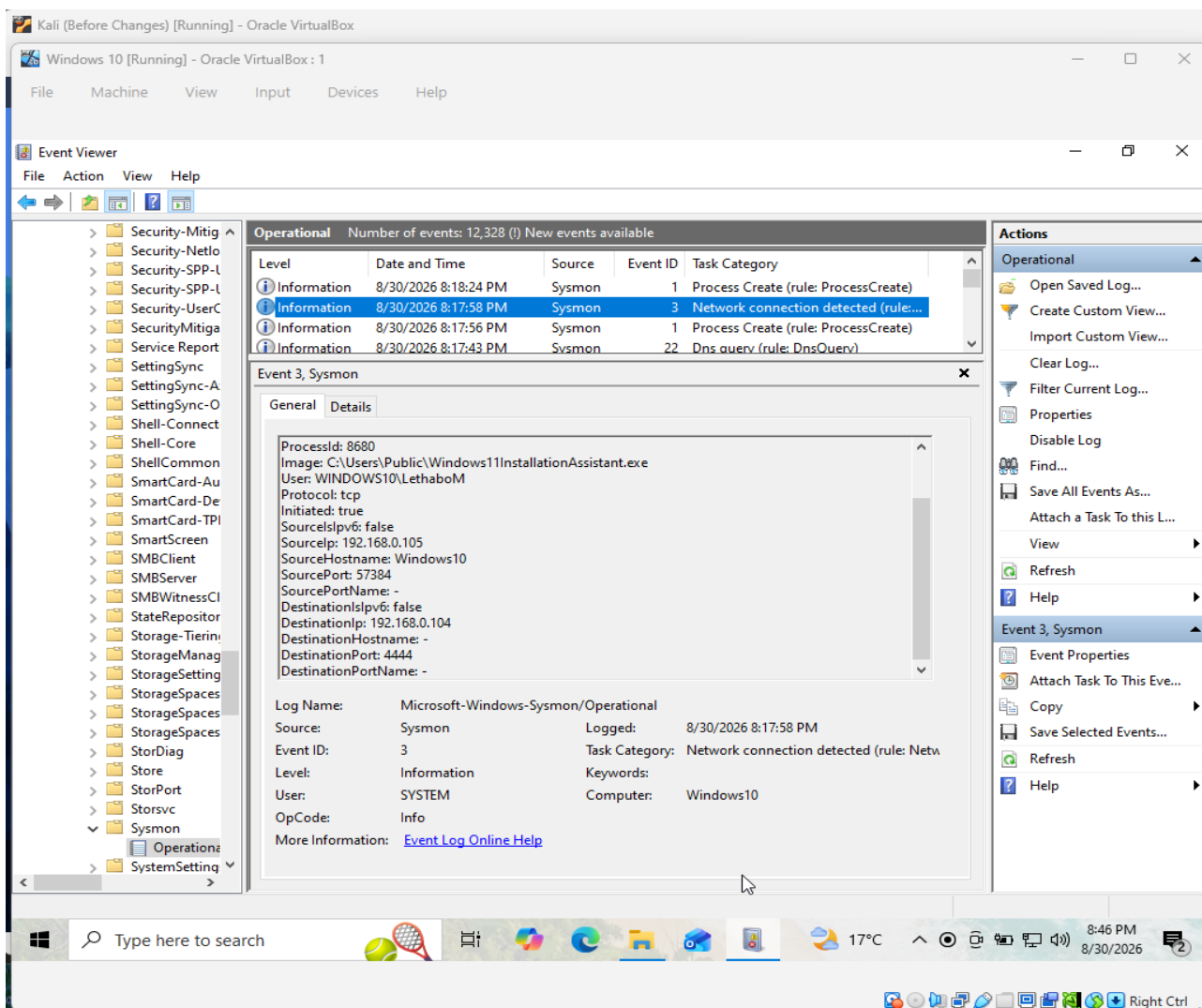
Destination Port: 4444

Protocol: TCP

Timestamp: 30 August 2026 08:17:58 PM

Evidence

Screenshot 02 — Sysmon Event ID 3



Finding

The suspicious executable established an outbound TCP connection to the controlled Kali system.

This provided additional context to the process-execution evidence.

The original lab identifies Sysmon Event ID 3, the destination IP, destination port and executable path as key evidence for this stage.

9. Persistence Investigation

The next stage of the investigation focused on persistence.

A scheduled task named:

WindowsUpdateHelper

was created.

The task was configured to execute the suspicious executable when a user logged on and to run under the SYSTEM account.

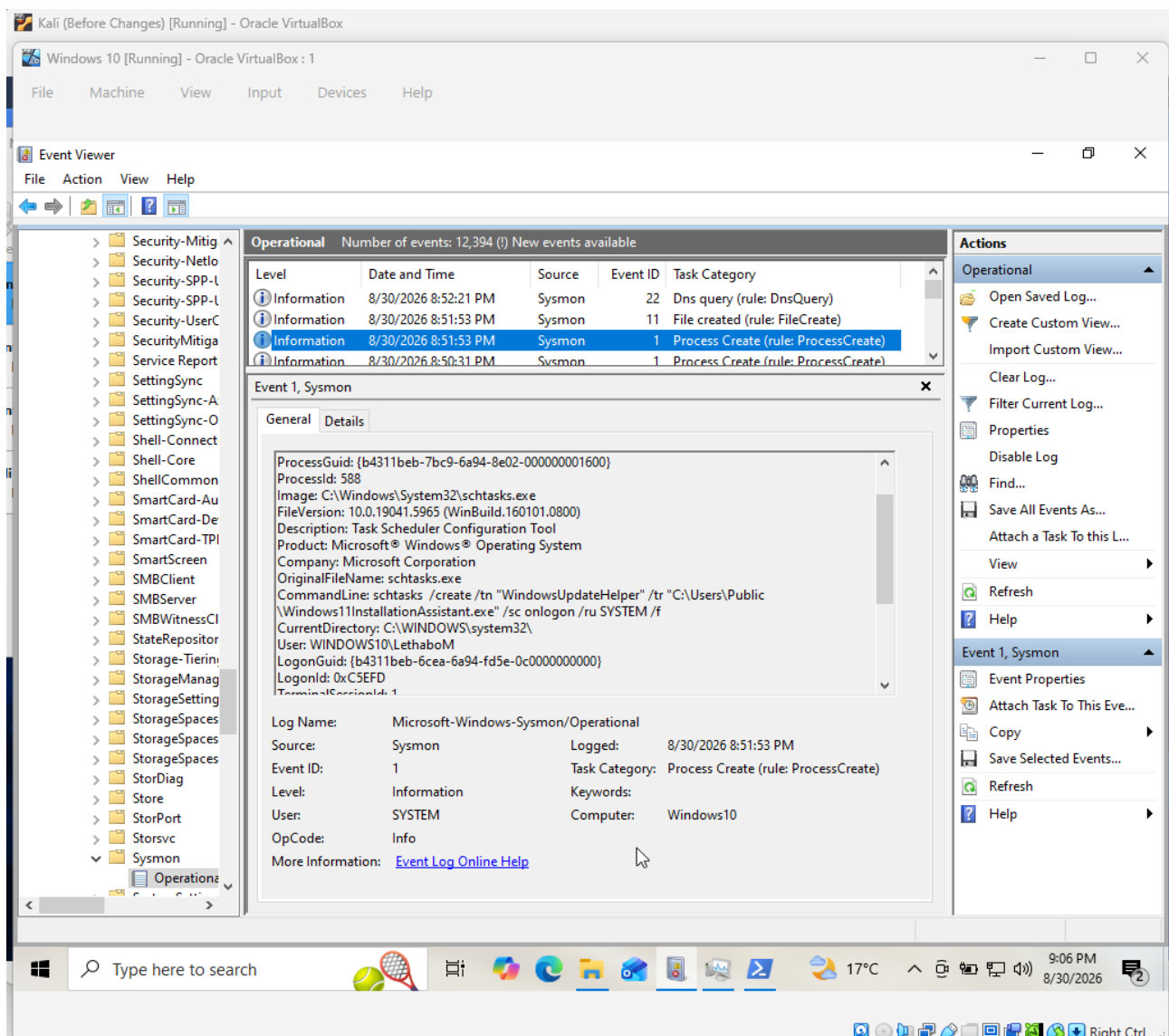
The important characteristics were:

Attribute	Finding
Task Name	WindowsUpdateHelper
Schedule Type	At logon time
Task To Run	Windows11InstallationAssistant.exe
Run As User	SYSTEM
Purpose	Persistence

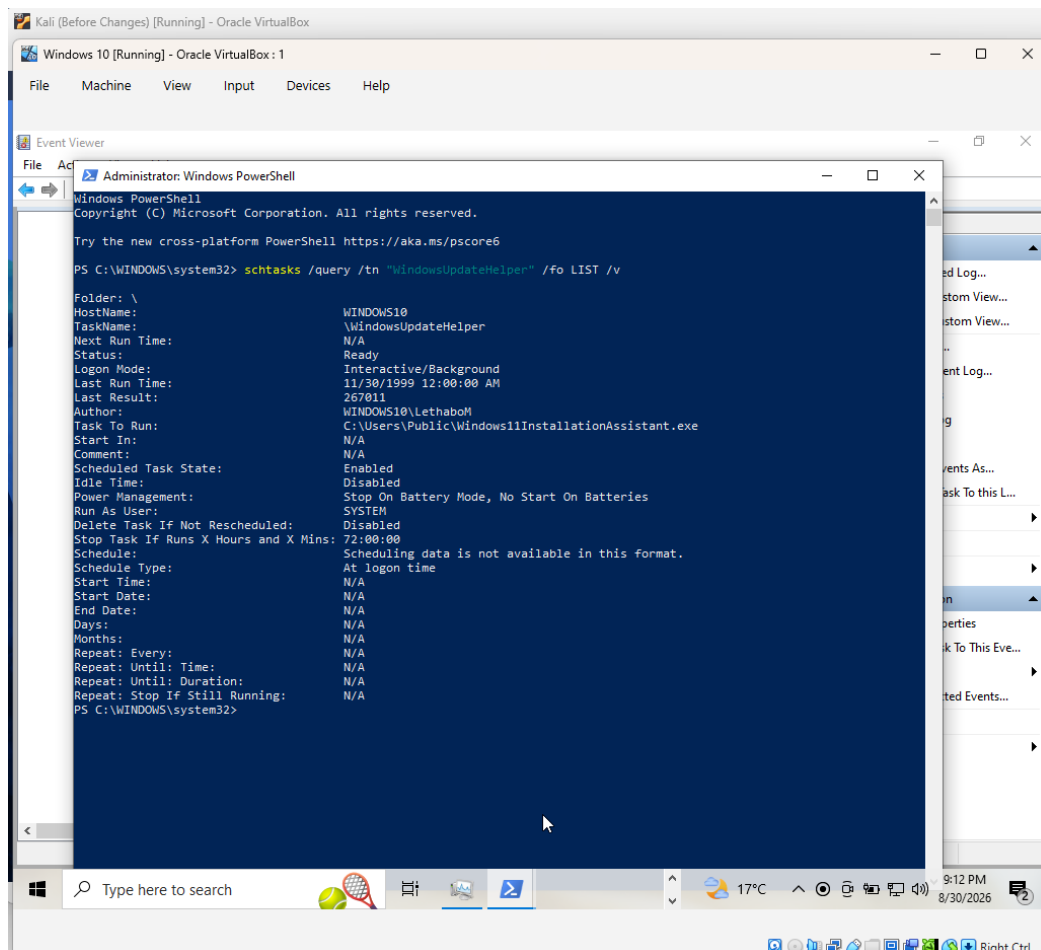
This was significant because the activity demonstrated an attempt to maintain execution beyond the initial compromise.

Evidence

Screenshot 04 — Sysmon Event ID 1



Screenshot 05 — Powershell Query WindowsUpdateHelper



11. Windows Security Event 4698

I then investigated **Windows Security Event ID 4698 — Scheduled Task Created**.

The event was reviewed to determine:

- Task name
- Task configuration
- Executable being launched
- Account associated with the task
- Creation timestamp

Expected evidence:

Event ID: 4698

Task Name: \WindowsUpdateHelper

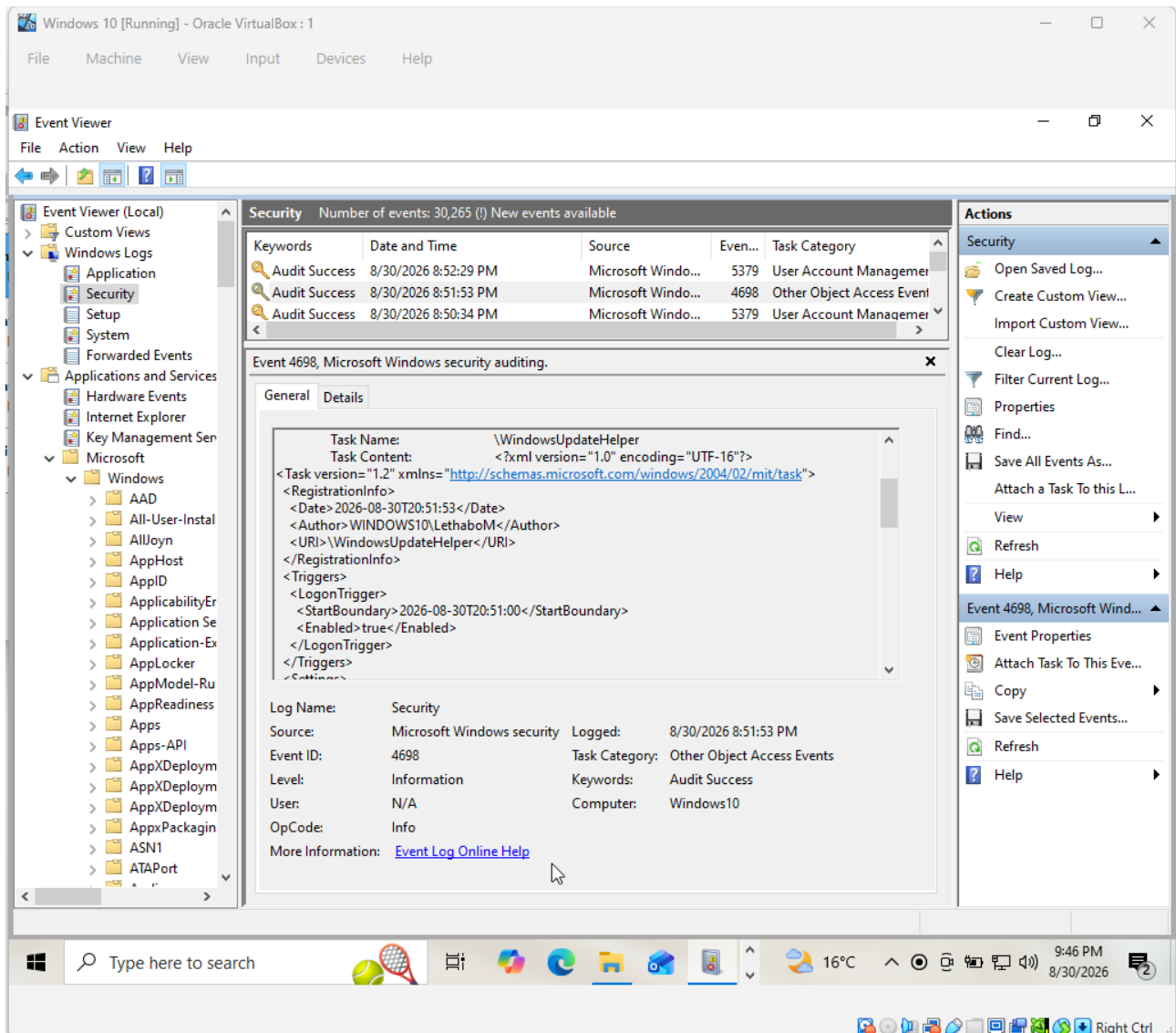
Exec: <Command>C:\Users\Public\Windows11InstallationAssistant.exe</Command>

RunLevel: LeastPrivilege

Timestamp: 30 August 2026 08:51:53 PM

Evidence

Screenshot 06 — Windows Security Event ID 4698



Finding

The event confirmed that the scheduled task had been created on the Windows endpoint.

The task configuration provided additional evidence of persistence.

12. Incident Timeline

The events were correlated into a single timeline.

Time	Event	Evidence
30 August 2026 08:17:56 PM	Windows11InstallationAssistant.exe executed	Sysmon EID 1
30 August 2026 08:17:58 PM	Outbound connection established	Sysmon EID 3
30 August 2026 08:51:53 PM	schtasks.exe executed	Sysmon EID 1
30 August 2026 08:51:53 PM	WindowsUpdateHelper created	Security EID 4698

The purpose of the timeline was to demonstrate how apparently separate events could be correlated into a single attack sequence.

The lab's intended sequence is executable execution → outbound connection → schtasks.exe → scheduled-task creation.

13. Indicators of Compromise

The following indicators were identified during the investigation.

File IOC

C:\Users\Public\Windows11InstallationAssistant.exe

Network IOC

192.168.0.104:4444

Persistence IOC

\WindowsUpdateHelper

Behavioral Indicators

- Suspicious executable execution
- Outbound network connection
- schtasks.exe execution
- Scheduled-task creation
- SYSTEM-level execution
- Persistence triggered by user logon

14. MITRE ATT&CK Mapping

The activity was mapped to the following MITRE ATT&CK techniques:

Technique	Description	Observed Activity
T1059.003	Windows Command Shell	Command execution
T1053.005	Scheduled Task/Job	Persistence
T1071.001	Web Protocols	HTTP delivery

15. Investigation Findings

The investigation established the following:

Finding 1 — Suspicious execution

A suspicious executable located in the Public directory was executed.

Finding 2 — Network activity

The executable established an outbound TCP connection to the controlled system.

Finding 3 — Persistence

A scheduled task named `WindowsUpdateHelper` was created.

Finding 4 — Process correlation

The suspicious executable was associated with the execution of `schtasks.exe`.

Finding 5 — Privileged persistence

The scheduled task was configured to execute under the `SYSTEM` account.

16. Verdict

TRUE POSITIVE — CONFIRMED MALICIOUS ACTIVITY

The combined evidence indicates activity consistent with post-exploitation persistence.

The conclusion was not based on a single event.

Instead, multiple independent pieces of telemetry were correlated:

Process Execution

+

Network Connection

+

Scheduled Task Creation

+

SYSTEM Execution

=

Confirmed Suspicious Activity

17. Incident Response Recommendations

If this were a production endpoint rather than an isolated lab, my immediate response would be:

Containment

1. Isolate the affected endpoint.
2. Prevent further communication with the suspicious destination.

Investigation

3. Review PowerShell activity.
4. Review Windows authentication events.
5. Investigate possible lateral movement.
6. Determine whether additional endpoints were affected.

Eradication

7. Remove the malicious scheduled task.
8. Remove the suspicious executable.
9. Search for additional persistence mechanisms.

Recovery

10. Restore the endpoint to a trusted state where appropriate.
11. Continue monitoring for recurrence.

Documentation

12. Preserve relevant evidence.
13. Record timestamps and IOCs.
14. Document all response actions.

These actions are consistent with the remediation and investigation recommendations in the lab.

18. Lessons Learned

This lab demonstrated several important incident-response concepts.

Event correlation

Individual events can be difficult to interpret in isolation. Correlating them provides a much clearer picture of the incident.

Process relationships

Parent-child process relationships can help establish how suspicious activity originated.

Endpoint telemetry

Sysmon provides valuable visibility into process and network activity that may not be obvious from standard Windows logs.

Persistence detection

Finding the initial executable is not enough. An investigator must determine whether the attacker established persistence.

Timeline analysis

Accurate timestamps allow an investigator to reconstruct the sequence of events.

Documentation

A technical investigation needs to be converted into a clear incident record that another analyst or client can understand.